LSD in the Treatment of Alcoholism

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Ninety-five alcoholic patients took part in a single-blind study of the efficacy of LSD treatment, which featured four treatment groups: LSD given with and without a therapist present, sodium amobarbital-methamphetamine hydrochloride given with a therapist present, and routine clinic care. At one-year follow-up, with 87 percent of the patients reporting, all groups showed significant improvement in the areas of drinking and employment, but there was no significant difference between groups on any improvement criterion measure. This study thus lends no support to the claims made for the efficacy of LSD treatment in alcoholism.

THE 20-YEAR USE of LSD in psychiatric disorder has been marked by a plethora of literary and clinical reports ranging from the mystical to the practical. In the realm of therapy, claims for the effectiveness of LSD treatment in many types of psychiatric disorder have been made, but few reports have been noted for their scientific adequacy. For instance, in the therapy of the neuroses there is only one well-controlled trial reported, and this notes lack of significant improvement with LSD in comparison with more orthodox therapies(12). In the therapy of alcoholism a number of studies largely from Saskatchewan reported major success with a single massive dose of LSD(2, 6, 10, 11, 16). However, as Smart and Storm have pointed out in their critical review(14), these studies were characterized by inadequate controls, few consistently applied pretreatment and post-treatment criteria, and poor follow-up.

Only two controlled evaluations of LSD in the treatment of alcoholism have been published.[1] These reports, by VanDusen and associates(18) and Smart(15), failed to show that the LSD experience, as described by them, was an effective adjunct to the clinical treatment of alcoholism. However, response to LSD is known to be highly dependent on such factors as patient expectation, setting, dosage, and therapist rapport. For this reason the results of these two controlled studies could only be assessed within the limitations imposed by their experimental settings, and it was considered of practical value to undertake a further LSD study with alcoholics, utilizing completely different experimental conditions and methodology.

Methodology

The London Clinic of the Alcoholism and Drug Addiction Research Foundation, Ontario, established in 1954, uses primarily an outpatient treatment approach with facilities for inpatient care available if necessary. Multiple professional disciplines are represented, with the greater part of therapy being carried out by social work staff under medical and psychiatric supervision. Strong emphasis is placed on an atmosphere of free discussion and in-service education. Teaching commitments to medical, psychology, and social work students are heavy, and research is encouraged. Approximately 40 percent of the patients are referred by a physician or an agency, and most of the rest

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[1] Since this paper was written a third report has appeared in the literature(9).

are self-referred. They are drawn largely from a lower- to middle-class population, with an intake of approximately 220 a year.

The author commenced the use of LSD in January 1965, gaining experience in its use in a prestudy phase by conducting 50 LSD sessions with a patient group with diagnoses of neurosis, alcoholism, and personality disorders; dosages of 50 to 3,000 μg. were used. A technique of split dosage was developed in order to achieve an optimal effect for the individual patient. Most clinic treatment staff, including the author, elected to take the drug under hospital conditions prior to the commencement of the study itself. All were positive in their reports on the experience, finding it personally meaningful and helpful. Intake to the scheme ensued from March 1965 to June 1966. A brief description of the procedure follows.

At the patient's initial approach to the clinic, in person or by telephone, he made contact with the clinic secretary—a person skilled in her ability to encourage interaction and set the patient at ease. At this contact, the necessity of attending six hours of initial intake documentation and interviews was explained to the patient and an appointment date given. Acutely ill patients were always given initial medical assistance until their scheduled appointment date. Since delay in intake might constitute a selection factor in that only the more highly motivated patients might return, this delay period was kept to less than 14 days. At intake date the patients underwent a six-hour intake procedure with standardized quantifiable documentation covering background history, alcoholic history, and present social, domestic, and interpersonal relationships. They were then assigned in rotation to therapists. They also received psychological testing—Differential Personality Inventory(5), Quick (I.Q.) Test(1), and Hidden Figures Test(4), as well as physical and psychiatric examinations.

Factors limiting a patient's inclusion in the scheme were as follows: The patient should be between 20 and 60, should have no severe liver, cardiovascular, or CNS disease, and should not be psychotic or have a history of previous psychotic breaks. He should admit to an alcohol problem and should not have had more than one interview at the clinic within the last five years. If he was found acceptable, he was then asked during psychiatric examination whether he would agree to a 24-hour admission to a neighboring general hospital psychiatric unit within the next few weeks. He was told that the purpose of this was to administer one of a number of drugs to help him understand himself and his drinking behavior better and to aid the treatment process. If he agreed, he was then randomly assigned to one of four groups.

1. (23 patients). The hospitalized patient was given 300 μg. of LSD in 10 ml. distilled water intravenously. One hour later another intravenous injection was given, the dosage dependent on clinical assessment. A nurse was present throughout the experience, but there was no therapist present.

2. (25 patients). The hospitalized patient was given 300 μg. of LSD in 10 ml. distilled water intravenously. One hour later another intravenous injection was given, the dosage dependent on clinical assessment. A nurse was present throughout with patient's therapist.

3. (22 patients). The hospitalized patient was given 3¾ gr. of sodium amobarbital (Sodium Amytal) and 30 mg. of methamphetamine hydrochloride (Methedrine) in 10 ml. distilled water. One hour later another intravenous injection was given, the dosage dependent on clinical assessment. A nurse was present throughout with patient's therapist.

4. (25 patients). There was routine clinic care with individual, group, and milieu therapy as determined by patient's therapist. There was no hospitalization for a special drug experience, the patient being told that, from analysis of his intake documentation, this was now felt to be unnecessary.

During all hospital experience the psychiatrist held overall responsibility for the patient, but his activity varied with the patient's group. In group 1, apart from clinical examinations and drug administration, the patient was left with a nurse in constant attendance for at least five hours or as long as necessary. She gave supportive nursing care but minimized verbal interaction.

In groups 2 and 3 the patient's therapist and psychiatrist acted as co-therapists. The therapist was with the patient for at least a four-hour period of active interviewing. The psychiatrist did not interact actively with the patient but was present intermittently to advise and to assist in standardizing the interview technique. Interaction by the therapist was of a warm, supportive nature, aiding the patient to focus particularly on current problems and interpersonal relationships. However, this emphasis did not exclude utilization of such occurrences as abreaction or transcendental experiences. Since many previous studies and our experience have emphasized that the LSD effect was so distinctive that the therapist could not fail to be aware of its use, this study was planned to be single blind for the patient.

In all hospital groups, the patient was admitted on the morning of therapy, he was to have had no phenothiazines in the last 24 hours and no MAO inhibitors in the preceding 14 days. A voice-actuated tape recording was taken and used for further analysis and supervision of interview techniques. The room used for the experience had simple furnishings and was quiet, shaded, and restful. A Posey restraint belt was used around the patient's waist so that he could move around in bed but could not leave it. He was reassured as to the use of the belt and thereafter generally accepted it without question. About six hours after the first intravenous injection, chlorpromazine, 200 mg., and a combination of amobarbital sodium and secobarbital sodium (Tuinal), 1½ gr. orally, were given. This was repeated for night sedation. On the following day the patient was questioned by the psychiatrist concerning his experience, and he completed standard forms, including the Linton and Langs Questionnaire(8). His therapist also interviewed him and arranged further contacts at the clinic as indicated.

Follow-up as close to the 12-month mark as possible following the patient's entry into the scheme was made by independent trained observers who were unaware of the original group to which the patient belonged. A follow-up was also made on those patients who refused the hospital scheme but were otherwise found suitable for it.

TABLE 1

Stages in Reduction to Final Scheme Size

NUMBER OF PATIENTS	STAGE	REDUCTION
337	Original clinic contact	74
263	Attendance at initial intake procedure	66
197	Completion of intake procedures	42
155	Suitable subjects for hospital experience	60
95	Agreement to and completion of scheme	

Follow-up questionnaires were phrased and scored in a similar manner to initial intake questionnaires to allow direct comparison. A two-year follow-up is now in progress.

Results

Patient Intake

Selection processes entering into the formation of final patient treatment groups are necessarily of importance and are detailed in table 1. The 74 dropouts prior to intake date were incompletely documented, but at least 44 percent were known to be of "skid row" background and were perhaps inadequately motivated to wish to spend the required six-hour intake time. The 66 dropouts during the intake procedure were primarily patients who came seeking acute emergency care only and had no interest in continued contact. The next reduction of 42 patients constitutes the group rejected on medical, psychiatric, or other grounds inherent in the scheme. Details are given in table 2. The final reduction group of 60 patients refused the hospital experience but were otherwise found suitable for it.

TABLE 2

Reasons for Rejection from Scheme

REASON	NUMBER REJECTED
Younger than 20 or older than 60	8
Severe liver, cardiovascular, or CNS disease	6
Psychotic, prepsychotic, or mental defect	5
Nonadmission of alcohol problem	6
Former patient with more than one interview in the past five years	8
Patient assigned to social work student	9
Total	42

Chi-square analysis of the final 95 patients in groups 1 to 4 regarding age, sex, referral source, severity of drinking, marital status, education, type and length of employment, and accommodation demonstrated the lack of any significant ($p = .05$ level) difference between groups. Thus, in terms of demographic variables, the final group was a homogeneous one, and table 3 indicates some of these parameters taken across all groups.

Eighty-seven percent of the subjects in groups 1 to 4 were successfully followed up and their responses to questionnaires corroborated where possible by information from relatives. In addition, 58 percent of the refusal group were successfully contacted for follow-up. Responses were compared for pre- and post-differences in the areas of: 1) drinking—abstinence, drinking practices (frequency, length of benders, etc.), consequences of drinking (loss of jobs and friends, physical complications), and attitudes toward drinking (recognition and acceptance of need for sobriety); 2) employment; 3) accommodation; 4) social relationships; and 5) marital status. An analysis of variance-repeated measures design was used to assess these pre- and post-measures.

Table 4 gives the result of this analysis across criterion measures. A statistically significant improvement ($p < .01$) occurred across all four groups on most drinking indices. Improvement approaching significance ($p < .10$) occurred in employment and "attitudes to drinking" data. However, there were no significant differences between groups; i.e., overall improvement occurred, but the use of drug therapy in the hospital did not confer special advantages over routine clinical therapy. Pretreatment and posttreatment measures of marital status, accommodation, and social relationships showed no changes approaching significance. The number of patient contacts with the clinic from entry into therapy to termination also did not differ significantly between groups (16.11 ± SD 11.07 contacts).

TABLE 3
Grouped Demographic Characteristics

CHARACTERISTIC	MEAN
Age	40.90 ± SD 9.95
Sex	
Male	87
Female	8
Social class	3.53 ± SD 0.93
Years of problem drinking	8.15 ± SD 7.68
IQ	99.31 ± SD 15.73
Social stability (Straus-Bacon[17])	3.14 ± SD 1.20
Years of education	9.95 ± SD 2.28

TABLE 4
Analysis of Variance-Repeated Measures

CRITERION	PRE-, POST-DIFFERENCES TAKEN ACROSS ALL TREATMENT GROUPS	DIFFERENCES BETWEEN TREATMENT GROUPS
Abstinence	$p < 0.01$	NS
Drinking practices	$p < 0.01$	NS
Consequences of drinking	$p < 0.01$	NS
Attitudes toward drinking	$0.10 > p > 0.05$	NS
Employment	$0.10 > p > 0.05$	NS
Accommodation	NS	NS
Social relations	NS	NS
Marital status	NS	NS

Dosage

The method of fractionation of dosage was found to be effective. Intravenously, a definite LSD effect was usually experienced in five to 15 minutes. However, if there were few perceptual changes or autonomic manifestations within 45 minutes, dosage was boosted, usually from 300 to 500 μg. In all cases this produced a significant drug effect. The optimum point for LSD therapy was considered to be at a level where the patient was fully oriented and still capable of meaningful interaction with the therapist. He usually experienced considerable subjective alteration of perceptual, cognitive, and affective levels of functioning with much symbolic imagery. In the case of sodium amobarbital-methamphetamine hydrochloride therapy, the patient remained fully oriented throughout, with pressure of speech and marked affect. Table 5 indicates the level of medication used.

TABLE 5
Drug Dosage

BASIC	ADDITIONAL
LSD, 300 μg.	264.3 ± SD 89.5
Sodium amobarbital (Sodium Amytal), 3¾ gr.	2.67 ± SD 1.81
Methamphetamine hydrochloride (Methedrine), 30 mg.	9.05 ± SD 10.98

Discussion

Results indicate that LSD in this setting conferred no evident advantages over more conventional modes of therapy. Thus, corroboration under different experimental conditions of the findings of two previous controlled studies in the literature lends further weight to the rejection of LSD as a significant drug in the treatment of alcoholism. In particular, this study was designed to meet possible criticism of the Toronto LSD study(15) in two areas:

1. The patient was not supervised in LSD therapy by his regular therapist but by a comparative stranger. Because of the importance of "set" in the LSD experience, it was possible that the absence of the usual therapist might have influenced the experience negatively. Care was taken in our study to ensure good rapport and close contact between patient and therapist before, during, and after the experience, but this feature still led to no apparent long-term benefit. It had also been hypothesized that an intense affective hospital experience early in the patient's contact with the clinic might facilitate positive transference and cement patient-therapist rapport. However, as measured by numbers of contacts by a patient with his therapist, no difference between groups could be established.

2. Instead of the high fixed dosage of the Toronto study, a flexible dosage was used, regulated to the patient's needs. Again, no apparent increase in efficacy of treatment occurred.

Intrinsic Action of LSD

There has been uncertainty as to whether LSD has an effect per se on personality modification or whether a therapist is necessary to the experience. Kalinowsky(7) writes: "The question remains whether LSD is a treatment in itself or whether it only facilitates psychotherapy in the same sense as Sodium Amytal. . . ." Sargent and Slater(13) write in the same vein. In addition, other nontherapy studies report that "normal" subjects report themselves capable of greater insight and enhanced enjoyment of life after the experience(3).

Group 1 (LSD given without a therapist present) was included in the study to clarify the question of how important a therapist was in the experience. As part of documentation following his drug experience, the patient reported on various subjective parameters relating to the occasion. These will be fully reviewed in a subsequent paper, but relevant to this discussion is the finding that between groups 1 and 2 there appeared to be no essential differences in affective, cognitive, and perceptual experiences under the drug. In addition, groups 1, 2, 3, i.e., with or without therapist, reported either no change after the experience or, more usually, some degree of lessening of tension, depression, irritability, and isolation. In addition, patients usually felt more optimistic about the future and believed they had increased insight. Where the therapist was present he was, however, usually felt to have been of great assistance, mainly as a supportive agent but also in his capacity of helping to guide the experience toward significant material.

The tentative conclusion may be drawn that the patient was able to obtain short-term benefit in two ways: by the support and direction of the therapist who helped him through often frightening intense drug experiences and by some nonspecific effect of the hospital drug experience. This action may lie in the strong support of extra attention and hospitalization or in a drug effect common to LSD and sodium amobarbital-methamphetamine hydrochloride. Since both of these drugs induce intense affective release, it is possible that some neurophysiological mechanism resulting from strong affective discharge produces the marked subjective improvement noted.

"Set" Toward LSD

Apart from two ex-mental-hospital patients who were more knowledgeable regarding treatment, apparently no patient was

aware of the possibility that he might be receiving LSD and did not even question this likelihood. This situation seems novel in the literature since most patients are surrounded and even deliberately given a "set" about LSD before their experience—and thus a stereotyped response might well be expected. In this study suggestibility effects were minimal: When the subject experienced perceptual and other changes, he was reassured that this was an expected part of drug action, and he was encouraged to continue to think about his drinking problems and life situation. Under these circumstances, it is of interest that only three out of 48 patients experienced definite transcendental effects in the usual sense of union with the universe or God. However, many did experience a sense of extreme closeness and understanding, sometimes spontaneously referred to as compassion, toward those in their immediate environment and in fantasy toward their family and other associates. This affective state occurred so frequently that, considering the absence of any patient expectation of drug effect, one is led to the conclusion that this may be a basic affective change specific to LSD. It is possible that the so-called transcendental state may be a combination of this basic affect with intellectual and mystic overtones induced by therapist bias and the patient's knowledge of the literature.

Summary

In an alcoholism clinic setting, a single-blind study of LSD in the treatment of alcoholism was undertaken. Ninety-five patients were randomly assigned to four groups. These groups were: 1) LSD given without a therapist present; 2) LSD given with a therapist present; 3) sodium amobarbital-methamphetamine hydrochloride given with a therapist present; and 4) routine clinic care.

At one-year follow-up, 87 percent of the patients were contacted. All groups had improved significantly across drinking and employment indices, but there was no significant difference between groups on any improvement criterion measure. This study therefore lends no support to the claims made for LSD in the treatment of alcoholism.

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DISCUSSION

JACK H. MENDELSON, M.D. (Chevy Chase, Md.).—This study represents an objective attempt to assess the efficacy of LSD treatment of alcoholics. Dr. Johnson's work is characterized by a high degree of methodological sophistication coupled with a recognition that the empirical value of drug therapies in any behavioral disorder is difficult to demonstrate in tightly controlled experimental studies. Dr. Johnson is quite correct in pointing out that most of the attempts that have been made to evaluate LSD treatment in alcoholism lack adequate controls and explicit criteria for long-term evaluation.

The selection of patient groups for this study is well described, and given the limitations of matching subjects for outpatient studies in alcohol-related areas, the subject selection and grouping have been carried out reasonably well. The experimental design is good. The author has wisely included the use of non-LSD medication given intravenously as well as no medication. He has also wisely incorporated the use of a behavioral end point for determining dosage in one of the patient groups rather than only utilizing a fixed dose throughout all patient groups.

The findings of this study highlight the lack of efficacy of LSD treatment as carried out under the conditions described. Moreover, these studies show that the presence or absence of a therapist during the LSD session makes little difference in eventual outcome. I think this was an important control to incorporate since there have been a variety of quasi-mystical reports concerning the therapist's importance and role during LSD administration. This was an important study to have carried out, but I find it surprising that one would entertain the hypothesis that a single or short-term exposure to LSD or any drug would produce significant therapeutic results for alcoholics. Alcoholism is a chronic disease, and it would be surprising to find that any one-shot approach would affect the course of a disorder influenced by multiple interacting biological, psychological, and sociocultural factors.

The negative results obtained by Dr. Johnson in studies carried out under carefully controlled conditions highlight the need for patience and balance before acceptance of panacea therapeutics. This statement should not be interpreted as a condolence for therapeutic nihilism but rather a plea for better design and more systematic studies of drug treatment for alcoholic patients.

The folly of mistaking a paradox for a discovery, a metaphor for a proof, a torrent of verbiage for a spring of capital truths, and oneself for an oracle, is inborn in us.

—PAUL VALÉRY